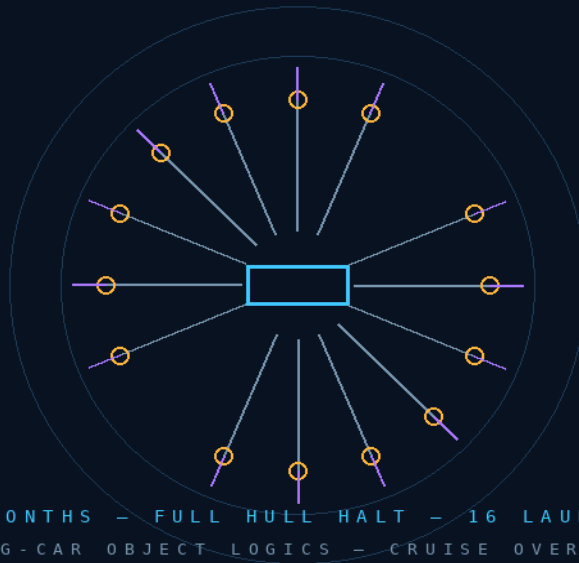


THE STARSHIP GALILEO
DEEP SPACE TRAVEL AND COLONIZATION
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS
PHASE 18

**16-POINT RADIAL SATELLITE ARRAY
MAPPING ASSEMBLIES**

Every 3 months — full hull halt — 16 launch as one
3P · 3S · 3T · 3B · 2F · 2A — self-driving-car object logics



EVERY 3 MONTHS — FULL HULL HALT — 16 LAUNCH AS ONE
SELF-DRIVING-CAR OBJECT LOGICS — CRUISE OVERRIDE ON MASS

GP-IM-018

Revision v1.1 — 21 August 2026

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VETTED: PASS — DISTRIBUTED RECONNAISSANCE ARRAY

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPhD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 18

PHASE 18: 16-POINT RADIAL SATELLITE ARRAY MAPPING ASSEMBLIES — STATUS: CURRENT. The ship's quarterly deep look: every 3 months, at a full hull halt, sixteen heavy mapping probes launch as one — 3 Port, 3 Starboard, 3 Top, 3 Bottom, 2 Forward, 2 Aft — and chart the whole sphere with self-driving-car object-detection logics, overriding cruise control to avoid mass. Sixteen viewpoints, parallax, redundancy: the unknown system gets measured from every axis at once. Vetted: PASS — distributed reconnaissance array, with the vet's terminology discipline seated in §2.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-018, v1.1 — 21 August 2026. The nineteenth manual of the program — 19 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the array law as seated (contents line and design body, verbatim) — §2 why it works (distributed sensing, graded) — §3 the system on the ship — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

VERSION NOTE (v1.0 — 20 August 2026): first build of this manual, on the Author's standing night-shift order (seated verbatim in sitting 499). The phase body stands exactly as seated in the master — split out of the combined PHASE 14, 15, 16, 17 & 18 cluster into its own standalone section in sitting 498, no wording changed; the manual adds the assembly truth around it.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Mounts a synchronized matrix of 16 independent, deployable satellite probes to execute real-time, multi-axis spatial charting of unknown planetary orbits and localized resource zones.'

THE SEATED DESIGN BODY (verbatim): '16-Point Radial Mapping: Every 3 months during a full hull halt, 16 heavy satellites launch simultaneously (3 Port, 3 Starboard, 3 Top, 3 Bottom, 2 Forward, 2 Aft—

excluding the horizontal plane). The array executes automated self-driving car object-detection radar logics, overriding cruise control to avoid masses.' *[AMENDED 21 AUGUST 2026 — the HALT LAW reread by its author: no full halt; a single carrier unit holding all sixteen launches rear of the ship, the ship slows slightly, deploy-scan-retrieve, and the ship catches up again — the §2 amendment governs operations.]*

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE DISTRIBUTED TRUTH: there is nothing exotic here — the vet's own words. A distributed sensor array provides multiple observation vectors, parallax, independent radar/optical observations, redundancy, and better spatial localisation than any one sensor. Sixteen probes see what one never can: the sphere, all at once, from outside the hull's own shadow.

THE TERMINOLOGY DISCIPLINE, SEATED: the word 'satellite' can drag an AI reader toward orbital insertion around a planet. These are DEPLOYABLE RECONNAISSANCE/MAPPING PROBES — they do not need to orbit anything to perform distributed mapping (the vet's correction, adopted). Where a probe is genuinely intended to enter orbit, the manual will say orbit; here, it does not.

THE HALT LAW: the launch happens at a full hull halt — the ship stops, the sixteen leave as one, the survey runs, the ship resumes. Simultaneity is the geometry: the whole sphere's baseline is captured in a single epoch, so every object in the chart shares one clock. *[AMENDED 21 AUGUST 2026 — the full halt is RETIRED as the deployment posture: the power cost of stopping is known and the design refuses to pay it. The carrier law below governs.]*

AMENDMENT — 21 AUGUST 2026 — THE HALT LAW, REREAD BY ITS AUTHOR (verbatim): "18 halt law, we know the power cost of stopping, so a single unit capable of holding all the sats will launch rear of the ship we slow slightly, whilst it deploys deploys the sats , scan retrieve back on the ship as we will have caught up again."

SEATED AS DEPLOYMENT LAW (audit reading, disclosed): (1) a single carrier unit, sized to hold all sixteen probes, launches REAR of the ship; (2) the ship slows SLIGHTLY — no halt; (3) the carrier deploys the sixteen, the survey runs, the probes are retrieved; (4) the carrier comes back aboard as the ship catches up to it — a rear-launched, decelerated carrier and a barely-slowed ship guarantees the rendezvous. The quarterly cadence and the one-epoch simultaneity stand; the SHIP no longer stops for them — the carrier takes the delta-v. ('deploys deploys' and 'sats' preserved verbatim above — typos ride the quote, the law rides clean.)

THE OBJECT LOGICS: automated self-driving-car object-detection radar logics, with the authority to override cruise control to avoid masses — the array is not a survey ornament; it is wired into the helm with override authority. Detection without action authority would be decoration.

§3 — THE SYSTEM ON THE SHIP

- THE CADENCE: every 3 months, standing — plus any halt the helm calls. The calendar is the minimum, not the ceiling. *[AMENDED 21 AUGUST 2026 — the cadence stands but the posture changed: no full hull halt; the rear-launched carrier runs deploy-scan-retrieve while the ship slows only slightly — §2.]*
- THE SIXTEEN: heavy probes, 3 Port / 3 Starboard / 3 Top / 3 Bottom / 2 Forward / 2 Aft — the horizontal plane deliberately excluded from the aft pair's sector, per the seated body.
- THE CHART: real-time multi-axis spatial charting of unknown planetary orbits and localized resource zones — the product is the map, and the map is property (§4).
- THE OVERRIDE: object-detection logics feed the cruise override — mass ahead stops the ship; the array's word is law at the helm.

INTEGRATION NOTE (audit, flagged): sixteen heavy probes plus recovery/stowage are mass and volume — stowage, launchers and the halt-ops procedure ride the hull architecture (GP-IM-006 v1.4) and the bench (§9).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE DRAGON-SCALE TRIGGER (seated doctrine): 'On detection of an incoming debris shower via Phase 15/18 scout arrays, the entire bank collapses into a solid disc wall ahead of the threat vector' — the array calls, the dragon scale answers (Phase 160).
- THE TRIANGULATION MANDATE (seated doctrine): 'Phase 18 16-point radial satellite arrays run unbroken triangular scans of the celestial horizon during transit' — the grained AI units process the changing angular delta between stars, logging cosmological expansion drift into the dynamic astrogation.
- THE ASTROMETRICS COPYRIGHT (seated law): 'Phase 18 16-point mapping data is the exclusive property of the Corporation. Public dissemination is restricted to low-resolution educational formats showing macro-scale planets and moons only.' — the map is the treasure; the treasure is guarded.
- THE PERIMETER PAIR (seated doctrine): with Phase 15 the array shares the early-warning perimeter and the scout-verification duty — 15 runs the standing forward cadence; 18 owns the deep quarterly sphere.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT, vet batch on phases 14-18), VERBATIM GRADE: ' PASS — distributed reconnaissance array.' Its findings, seated: 'There is nothing exotic here' — multiple vectors, parallax, independent observations, redundancy, better spatial localisation. Its one terminology problem

is seated as law of reading in §2: 'satellite' reads as deployable reconnaissance/mapping probes unless orbital insertion is specifically intended — 'they don't need to orbit anything to perform distributed mapping.' No flag against the concept; the implementation note ('deployed simultaneously around the vehicle') is the seated design body itself.

§6 — BUILD SEQUENCE

- STEP 1 — Build the probe: heavy reconnaissance/mapping unit — radar + optical heads, object-detection computer, comms back to the helm, propulsion sufficient for station-keeping on its survey point.
- STEP 2 — Build the sixteen as one lot: identical units, one maintenance line, one spares pool — 3P/3S/3T/3B/2F/2A assignment painted on, not designed in.
- STEP 3 — Fit the launchers and stowage: simultaneous release at the halt — the sixteen leave together or the epoch breaks.
- STEP 4 — Commission the object logics: the self-driving-car detection stack, the mass-avoid override path to cruise control, and the dead-man test — override fails safe, never fails silent.
- STEP 5 — Commission the chart pipeline: triangular scans, shared-epoch bookkeeping, the astrometrics vault (property law applies — §4), and the feed to the dragon-scale trigger.
- STEP 6 — Bench the survey figures (§9) and seat them; only then call the array operational.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 15 — the phantoms (GP-IM-015 v1.0): the 15/18 pair — the dragon-scale trigger and the early-warning perimeter are joint doctrine (§4).
- PHASE 160 — The Dragon Scale Kinetic Deflection Shield: the wall fires on 15/18 detection — the array's warning is the shield's trigger.
- PHASE 16 — the buoys (GP-IM-016 v1.0): the array's returns ride the highway home; the astrometrics vault syncs over it.
- PHASE 17 — the beacons (GP-IM-017 v1.0): the flush-optics tracking discipline the probes inherit; numbered, pinged, mapped.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): stowage, launchers and halt-ops seat in the hull architecture.
- PHASE 14 — the acoustic shields (GP-IM-014 v1.3): the former cluster mate — 14 answers vibration; 18 owns the sphere.

§8 — DO-NOT-BUILD

- NEVER call the probes 'satellites' in spec text — deployable reconnaissance/mapping probes; orbit only where orbit is meant (§2).
- NEVER survey on the move — the halt is law; a running launch breaks the shared epoch (§2).
- NEVER let the override fail silent — detection without action authority is decoration (§6, step 4).
- NEVER leak the chart — the astrometrics copyright is seated law; public gets low-resolution macro scale only (§4).

§9 — OWED

THE BENCH, OWED: probe unit mass, power, propulsion and endurance; survey standoff geometry per quarter; detection range and the override's decision budget (closing speed vs halt authority); launcher sync tolerance (how simultaneous is simultaneous); recovery/refurbish-or-replace cycle; chart data rates into the vault. All owed items ride the bench until spoken. *[PARTIALLY ANSWERED 21 AUGUST 2026 — the recovery cycle is now the carrier law: one carrier holds all sixteen, deploys, retrieves, and is caught up to. New owed items from the amendment: carrier sizing, stowage and launch rail for a sixteen-probe carrier; carrier propulsion/delta-v budget (it takes the stopping cost the ship refuses to pay); deploy/retrieve sequencing at survey standoff; rendezvous window and catch-up math. The rest ride.]*

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-018, v1.1 — CURRENT, seated law, master v2.1 (numerical print order, post-split). The nineteenth manual of the program — 19 of 290 delivered. Built 20 August 2026; v1.1 seated 21 August 2026 — the HALT LAW reread by its author: full halt retired, rear-launched carrier deployment seated.

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'. The phase's own chain, all seated in the master: the design body and Contents line from the July record; the naming batches of 12 August 2026 ('18: 16-Point Radial Satellite Array Mapping Assemblies'); the vet's PASS with the satellite-terminology correction; the split of 20 August 2026 that stood the phase alone (sitting 498, his order).

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete. Next version triggers: the §9 benches, halt-ops rulings, or his word. v1.1 (21 August 2026): the halt-law amendment seated at §1, §2 and §3; §9 gains the carrier benches.

END OF MANUAL — PHASE 18